

S1. Setup: several JEWEL-like jets per PYTHIA jet

Question. OT-FM (the deterministic OT-CFM translator) gives one output per PYTHIA jet. Can noise-driven conditional flow matching keep its JEWEL-like populations and input dependence, with sharp, different samples of each jet? Exploratory: the conditional distribution is the one the pairing defines, not a physical medium response.

Model (standard flow matching [Lipman et al., arXiv:2210.02747]; not DSBM, not an SDE bridge)

- $x_t = (1 - t)\epsilon + ty$, $u_t = y - \epsilon$, $\mathcal{L} = \langle (v_\theta(t, [x_t, c]) - u_t)^2 \rangle$; c : the PYTHIA jet, an unchanged second input channel; y : its paired JEWEL training jet; ϵ : Gaussian on the 53 cone towers
- sampling: fix c , draw ϵ , midpoint ODE with 128 calls; the same (c, ϵ) give the same jet
- **pairs** [Tong et al., arXiv:2302.00482]: in each batch of 256 PYTHIA and 256 JEWEL training jets a plan on the squared L^2 of the standardised $\log(E + 0.1)$ canvases, one target drawn per condition. ■ **hard**: exact OT. ■ **soft**: entropic OT, reg 7, i.e. a median of 4 targets per row (5-95%: 1.6-9.2), fixed on training jets before training
- equal in both: data, batches, times, noise, initialisation, the UVCGAN-S network (2 input channels), seed 0, 2 A6000 GPU-hours

The training pairs themselves (8192 pairs, before training)

	EMD	$r(p_T)$	$r(\text{girth})$	rms Δp_T
■ exact plan (hard)	0.235	0.17	0.71	11.8
■ entropic plan (soft)	0.240	0.16	0.70	11.9
random pairs	0.465	0.00	0.02	12.7
■ OT-FM output vs its input	0.041	0.94	0.99	6.6

r : correlation of a condition's and its target's observable. An OT partner shares part of the condition's width and almost none of its p_T .

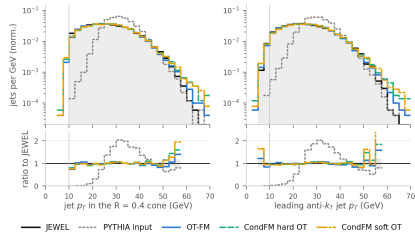
Checks

- path in the loss, pair indices, fixed-noise repeats: exact
- Gaussian control $y = c + 0.2\xi$: samples of a fixed c have mean within 0.013 of c , sd 0.196-0.208
- solver (validation): 128 and 256 calls from the same noise differ by 0.005 GeV per jet, two noises by 13.4 GeV: resolved at 128

S2. One sample per held-out PYTHIA jet: the population benchmark

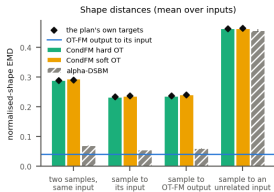
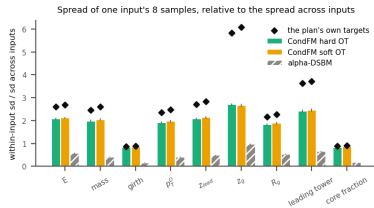
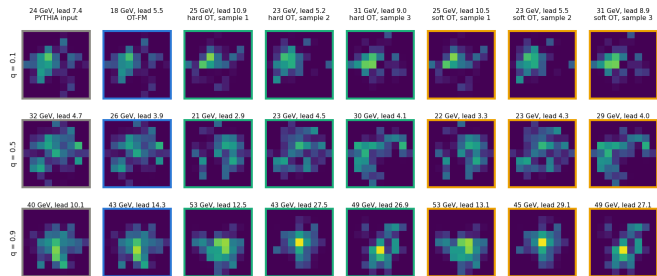
	distance to held-out JEWEL, W_1/σ							input dependence		
	p_T	mass	girth	p_T^D	z_{lead}	z_g	R_g	$r(p_T)$	$r(\text{girth})$	own
JEWEL vs JEWEL	0.011	0.014	0.009	0.018	0.019	0.012	0.008			
random JEWEL	0.019	0.024	0.011	0.010	0.011	0.005	0.010	0.01	-0.00	0.50
PYTHIA input	0.546	0.881	0.105	0.264	0.232	0.034	0.158	1.00	1.00	1.00
OT-FM	0.025	0.021	0.012	0.010	0.008	0.017	0.012	0.94	0.99	1.00
hard OT	0.055	0.055	0.017	0.015	0.012	0.006	0.013	0.19	0.72	0.92
soft OT	0.056	0.059	0.020	0.019	0.017	0.007	0.017	0.18	0.71	0.92
α -DSBM	0.256	0.325	0.056	0.112	0.099	0.036	0.073	0.80	0.98	1.00
CycleGAN 2 h	3.914	3.327	0.143	0.657	0.579	0.123	0.220	0.85	0.99	1.00

20k test jets each; W_1/σ after $p_T \geq 10$ GeV on every sample (bootstrap sd 0.004-0.010); own: share of outputs closer in shape to their input than to another jet. CycleGAN: the matched 2 hours.



- **Shapes** as close to JEWEL as **OT-FM's**, also at fixed p_T and for re-found jets
- p_T and **mass** 2-3 $\times$ **OT-FM's** distance (too many jets above 50 GeV): 2.7-2.9 sd (**hard**), mass 3.4 sd (**soft**). A second sampler seed moves every distance by ≤ 0.011
- **Single jets sharp**: occupancy, leading tower, core and soft energy as JEWEL's; 5% of samples in JEWEL's 5% tails; no flag
- **The input sets little of the output p_T** : $r = 0.19$ (**OT-FM** 0.94); median output 23 GeV for 20-25 GeV inputs, 30 GeV for 50-70 GeV inputs

S3. Eight samples per jet: how much they differ, and what they follow



- **Large, real diversity:** two samples of one jet differ by 13 GeV rms in p_T and 0.29 in shape EMD, more than a sample differs from its input (0.23; unrelated jets 0.47; **OT-FM's** change 0.04; solver 0.005 GeV)
- **Not only energy:** within one input p_T^D , z_{lead} , z_g , R_g spread as much as over all JEWEL jets, girth and core 65% as much; coherently (p_T with lead tower r 0.78)
- **The pairing's spread (♦):** the targets each plan assigns to the same inputs spread and correlate with them as the samples do (table)
- **Swap c at fixed ϵ :** the output follows the new jet's shape (93% closer to it than to the old input); the noise adds a shared p_T and width shift (r 0.39, 0.52)
- **hard = soft:** same input and noise: 0.012 in shape EMD, 0.8 GeV apart

$r(\text{sample, input})$	p_T	mass	girth	z_{lead}	core
— OT-FM	0.94	0.92	0.99	0.96	0.99
— hard OT	0.17	0.25	0.73	0.24	0.72
its pairs	0.16	0.25	0.72	0.24	0.71
— soft OT	0.16	0.24	0.72	0.22	0.71
its pairs	0.15	0.23	0.71	0.23	0.70

S4. Reading and cost

Pre-registered reading

	— hard OT	— soft OT
toward JEWEL (≥ 4 of 7, both joints)	yes (6 of 7)	yes (6 of 7)
none 3 sd worse than OT-FM	yes	no (mass)
input dependence beyond random	yes	yes
sharp samples (tails $\leq 10\%$)	yes (5%, 5%)	yes (5%, 6%)
solver resolved	yes	yes
diversity above numerical error	yes	yes
follows the condition (swap)	yes (66%)	yes (64%)
useful (all of the above)	yes	no

Cost (one A6000 each; training time; latency at batch 2000)

	GPU h	updates	per s	OT	calls	ms/jet
— OT-FM	2.00	123k	17.0	10%	128	6.58
— hard OT	2.00	122k	17.0	10%	128	6.60
— soft OT	2.00	121k	16.9	11%	128	6.67
α -DSBM	2.00	35k	—	—	30	1.55
CycleGAN 2 h	2.06	20k	—	—	1	0.05

- **By the letter hard** meets every item; **soft** misses one (mass 3.4 sd behind **OT-FM**).
- **What the samples are:** sharp JEWEL-like jets that share part of the input's width and core and little of its p_T . The model learned its pairing's conditional distribution faithfully; a minibatch-OT pair keeps $r(p_T) = 0.17$. **OT-FM**'s tight per-jet relation ($r = 0.94$) comes from the deterministic flow, not from the pairs.
- **More spread is not better here:** the spread is the pairing's, not a medium fluctuation; the entropic plan adds nothing measurable to the variety of the hard plan's partners.
- **Variability:** all spreads are sampling variability of one trained model per arm; a second sampler seed moves the benchmark by about one bootstrap sd; training-seed variability is not measured.
- **Not shown:** a physical medium response, a calibrated uncertainty, a per-jet truth. Keeping the input p_T would need a pairing that keeps it: a modelling choice unpaired data cannot make.